

CLI Commands

```
# Generate report → reports/unifi_report_YYYYMMDD_HHMMSS.html
python generate_report.py

python generate_report.py --open           # open in browser
python generate_report.py --config path.yaml # alt config
python generate_report.py --debug         # verbose output
python generate_report.py --version

# Generate demo report (dummy data, no controller needed)
python generate_demo_report.py --open
```

config.yaml

```
unifi:
  host:      "192.168.1.1" # controller IP or hostname
  port:      443           # 443 = UDM · 8443 = standalone
  username:  "admin"
  password:  "your-password"
  verify_ssl: false       # true if valid TLS cert
  # api_key: "your-key"   # optional — enables v1 API sections

report:
  site_name: "My Network"
  output_dir: "reports"
  timezone:  "America/Chicago"
  thresholds:
    ap_retry_pct: 5.0 # AP TX retry % flag
    ap_reboot_days: 7 # recent reboot window (days)
    switch_error_rate: 100 # port error count flag
```

Report Sections (top → bottom)

Section	Key Data Source
Recommendations	Computed from all sections
Gateway / WAN	stat/health · stat/device
WAN Utilization Chart	stat/report/hourly.site
Access Points	stat/device (type: uap)
Switches	stat/device (type: usw)
Firmware Compliance	stat/device (all types)
Traffic Analysis	stat/report/daily+hourly.user · rest/user
Top Bandwidth Consumers	stat/sta
Client Inventory	rest/user (all known + historical)
Wireless Networks	rest/wlanconf · stat/health
VPN	stat/health (vpn subsystem)
Networks & VLANs	rest/networkconf
Zone Firewall v1	integration/v1/firewall/policies+zones
DNS & Content Filtering v1	integration/v1/dns/policies
Legacy Firewall Rules	rest/firewallrule · firewallgroup · portforward
Event Timeline	v2/api/site/{site}/event

v1 sections require api_key in config (NA 10.1+) — hidden when absent.

Supported Controllers

Controller	Port	API Prefix
UDM / UDM-Pro / UDM-SE / UDM-Pro Max	443	/proxy/network/
Network Application (self-hosted)	8443	(none — direct)

Tested: Network Application 10.3.58 on UDM-SE firmware 5.0.16. v1 API available on both.

API Endpoints

LEGACY API — COOKIE SESSION

Endpoint	Returns
stat/health	WAN/LAN/WLAN status, client counts, speedtest
stat/device	All devices (APs, switches, gateways) + real-time stats
stat/sta	Connected clients — IP, MAC, session bandwidth
rest/user	All known clients incl. disconnected + fingerprinting
rest/wlanconf	SSID config — security, bands, guest flag
rest/networkconf	Networks, VLANs, subnets, DHCP
rest/firewallrule	Legacy user-defined firewall rules
rest/firewallgroup	Address & port groups
rest/alarm	Active unresolved alarms
rest/portforward	Port-forwarding / DNAT rules
v2/.../trafficroutes	Policy routing / WAN-selection rules
stat/report/hourly.site	WAN TX/RX bytes per hour (24 h)
stat/report/daily.user	Per-client TX/RX totals (24 h)
stat/report/hourly.user	Per-client TX/RX bytes per hour
v2/.../event	Recent controller events

V1 INTEGRATION API — X-API-KEY SESSION (NA 10.1+)

Endpoint	Returns
/integration/v1/sites	Site list with UUIDs (needed for all v1 calls)
.../firewall/policies	Zone-based firewall policies
.../firewall/zones	Zone definitions + network membership
.../dns/policies	DNS / content-filtering policies

UDM paths auto-prefixed with /proxy/network/. Legacy API is unofficial; v1 is the documented Ubiquiti API.

Auto-Generated Recommendations

Check	Config Key	Default	Priority
AP TX retry rate	ap_retry_pct	5.0%	Medium
AP recently rebooted	ap_reboot_days	7 days	High
AP offline	—	any	High
Switch port errors	switch_error_rate	100	Medium
Active alarms	—	any	High
WAN status not ok	—	any	High
WAN_IN accept, unrestricted source	—	any	High

Key Concepts

site slug vs site UUID

Legacy API uses slug (e.g. default). v1 API requires UUID — fetched via GET /v1/sites, matched on internalReference, then cached.

dev_cat

Integer fingerprint code from rest/user. Mapped to labels (Computer, Phone, IoT...) via _DEV_CAT_GROUP in generator.py for Traffic Analysis category bars.

stat/report POST quirks

Requires ms timestamps + "time" in attrs payload. NA 10.x returns MAC in "user" / "oid" field, not "mac".

SVG charts

WAN utilization and hourly activity charts are pre-computed server-side in Python — no client-side JS library. Coordinates passed as polyline strings to Jinja2.

JSON Snapshot

Compact reports/json/unifi_snapshot_*.json written alongside each HTML. Contains WAN, clients, firmware, firewall summary, top-20 traffic — excludes SVG paths.

Themes

Light · Dark Terminal · Slate & Amber · Deep Navy — CSS custom properties + data-theme on <html>. Choice and section collapse state saved in localStorage.

Troubleshooting

```
# SSL errors      set verify_ssl: false in config.yaml

# Section missing python generate_report.py --debug
                  look for WARNING ... 404 / 401 / returned empty

# v1 API → 401    api_key wrong or expired — regenerate in:
                  UniFi Network → Settings → System → Advanced → API

Keys

# Traffic Analysis stat/report requires active clients + NA 10.x
# shows no data   MAC in "user"/"oid" field on NA 10.x, not "mac"

# stat/dpi empty  Expected on NA 10.x — use dev_cat category
breakdown instead
```